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Support for Government Health Insurance Before and During COVID-19

The United States is one of the only developed nations that does not offer a form of universal government healthcare. However, the largest single source of health insurance in the United States are the government programs Medicaid and Medicare which together cover 35.8% of the insured population (KFF 2022). Considering that statistical research has been done linking the health outcomes of COVID-19 with Medicaid expansion, this research seeks to find the impact of the COVID-19 pandemic on the general support for a universal government health insurance program at the state level. In order to achieve this goal, this research uses the American National Election Studies (ANES) time series survey from 2016 and 2020 and a multilevel regression and poststratification (MRP) model to determine the individual support for government health insurance at the state level. Our hypothesis is that the COVID-19 pandemic, having a significant impact on the health and lifestyles of a cross-section of people in the US, increased the support for government health insurance from 2016 to 2020.

Literature Review

In 2019, over 29 million people were uninsured in the United States, representing more than 9 percent of the total population (KFF 2022). At the start of the pandemic there was widespread concern about the number of uninsured persons and how it would impact the healthcare system. Those without insurance were more likely to be essential workers, low income, and have unmanaged chronic disease (Wilper et al.). Furthermore, government and official government health guidance recommended that regular doctor appointments and even

treatment for less severe cases of COVID-19 be performed over telehealth visits with primary care physicians. However, those without health insurance were far less likely to have an existing relationship with a primary care physician. Additionally, the high cost of healthcare and the risk of infection dissuaded many uninsured people from seeking treatment for both COVID-19 and other conditions.

In March of 2010, President Barack Obama signed into law a sweeping piece of legislation formally known as The Patient Protection and Affordable Care Act (ACA). The major reform included, among other provisions, new regulations for insurance companies, the creation of exchanges where Americans can buy health insurance policies, and a large-scale expansion of coverage eligibility for the existing Medicaid system. Particularly, the law stipulated that Medicaid coverage would be expanded to all adults with income below 138% of the federal poverty level. However, due to Medicaid's management at the state level, not all of the states adopted the Medicaid expansion sponsored by the bill. 13 years since the passage of the law colloquially known as Obamacare, all but 10 states have adopted the expansion of Medicaid. Since the first lockdown of the coronavirus pandemic in March of 2020, five states have adopted and implemented the expansion of medicaid under the ACA.

In states that expanded medicaid coverage before the start of the pandemic, increased eligibility meant coverage for those who lost their jobs and incomes due to the recession related to the pandemic. Populations in states that had expanded Medicaid also benefited from changes to federal policy such as the Families First Coronavirus Response Act which required continuous coverage for those enrolled in Medicaid during the public health emergency. Therefore, we can expect that individuals who were on Medicaid or became eligible for Medicaid during the pandemic and enrolled would have recognized the importance of having access to healthcare

during the pandemic. The increased dependence on government funded healthcare programs such as Medicaid would have theoretically translated into increased support for government insurance programs.

Question of Interest

The COVID-19 pandemic increased demand for healthcare worldwide. As the pandemic spread, hospitalizations and severe infections increased concern over the cost of healthcare. The largest impact was on the uninsured populations who faced substantial costs associated with accessing health care. Additionally, as most insurance in the United States is obtained through employers, job loss due to the pandemic resulted in the loss of health insurance and necessitated that individuals search for other options such as private health insurance or medicaid. Therefore, we are interested in how the support for a government health insurance program changed from before the pandemic to during the pandemic.

Hypothesis

Our hypothesis is that state support for government health insurance increased during the pandemic from pre-pandemic. We expect to find a general increase in the mean support for government health insurance, as captured by our selected variable in the ANES, from 2016 to 2020.

Datasets

Due to ease of access, the ANES Time Series Cumulative Data File was used to gauge public opinion on health care. We chose to use the Cumulative Data File, versus individual year data files, to ensure uniformity between the years of interest. Despite our efforts for uniformity, it should be noted that live interviews in 2020 were conducted over video call, whereas in 2016 they were conducted in person. The ANES dataset is relatively large, with several variables and

years that are unnecessary for the scope of our research. Two subsets of the original dataset were created, one containing observations for the year 2016, and the other for the year 2020. In these subsets we selected our variables of interest including demographic variables that will be used for a poststratification and our predictor variable ‘VCF0806’. This variable originally was an ordinal variable ranging from 1 to 7, 1 referring to a respondent who believes there should be a government health insurance plan that covers all medical expenses and 7 indicating that the respondent believes all medical expenses should be paid by the individual. This variable was converted into a binary variable, respondents choosing 1, 2, or 3 recorded as a 1, and respondents choosing 4 through 7 became a 0.

Table 1 (see Appendix A) shows us the distribution of the ANES dataset sample. Here, we see that there is an important disbalance in the dataset. For example, 70% (2016) and 72% (2020) of the sample identified as White. According to the 2020 census, people who declared themselves as white made up 60% of the total population. Another example of imbalanced data is people who identify themselves as Hispanic, who are underrepresented in both measures and who, according to official census data (2020 census), make up 18% of the total population. This carries over when viewed on a state-by-state basis, and therefore, poststratification is necessary to achieve more reliable estimates at the state level.

Poststratification tables should contain information about the number of people in a population of interest that fall into a given combination of demographic-geographic factors (Lopez-Martin et al.). This data can be gathered from several sources, typically the Decennial Census Data, which we were unable to use directly due to a lack of resources and time constraints. The poststratification table was obtained through the public data file from a paper entitled “Multilevel Regression and Poststratification Case Studies” by Lopez-Martin et al.,

which provides guidance on how to perform an MRP as well as its limitations. We recoded relevant demographic variables in the ANES to fit with the demographic variables in the poststratification table. ‘VCF0101’, the age variable, was divided into six bins ranging from 18 to 29, 30 to 39, 40 to 49, 50 to 59, 60 to 69, and 70+. ‘VCF0110’, the education variable, was recoded into 4 levels referring to no high school, high school, some college, and 4-year college. ‘VCF0104’, the gender variable, was changed to have 0.5 refer to male and -0.5 refer to female, there was no other category for this variable. Lastly, ‘VCF0105b’ was our ethnicity variable, which was recoded from a numeric variable into a character variable with the categories White, Black, Hispanic and Other.

In addition to the poststratification table, Lopez-Martin et al. also used a dataframe called ‘statelevel_predictors_df’ which contained regional and state variables as well as the republican vote share by state. We also included this in our model as a group-level predictor to account for potential unexplained variation by state and to produce precise estimates (Lopez-Martin et al. 2022). DC and the US Territories were removed from our analysis.

Methodology

We will implement a multilevel regression poststratification (MRP) strategy. We deployed this method because we can use survey data with individual-level observations and unrepresentative samples and transform our outcome into a state-level estimate. In that sense, we can use the multilevel regression to predict the outcome of interest using our individual and group variables. Then, we can implement the post-stratification using the population size from the census or other data to get the population estimates.

Furthermore, our model includes the effects of varying age, ethnicity, education, and state and Republican vote shares for each respective year. For this case, the only group-level variable

we will use is Republican share votes; all the other variables are taken as individual variables. The outcome variable is the belief that there should be a government health insurance plan that covers all medical expenses. It is a dummy variable, making our model a logistic regression; it estimates the probability of an event occurring based on a given dataset of independent variables.

The multilevel regression that we used is the following:

$$Pr(y_i = 1) = \text{logit}^{-1}(\alpha_{[i]}^{states} + \alpha_{[i]}^{age} + \alpha_{[i]}^{eth} + \alpha_{[i]}^{educ} + \alpha_{[i]}^{educ*age} + \alpha_{[i]}^{educ*eth} + \alpha_{[i]}^{male*eth} + \beta^{repvotes} \cdot RepVotes)$$

Once we implement the model over our survey data, we would obtain our estimates on an individual level, so the next step is the implementation of the poststratification, in this way we will go from this model to a subnational estimate. To do so, we will weigh the model predictions for the different subgroups by the actual frequency of the subgroups using our census data (Lopez-Martin et al. 2022).

Analysis

We will start our analysis by interpreting the multilevel regression for both years (2016 and 2020). For our purpose, the standard error estimates' interpretation is not the subject of this study. However, the tables show each standard deviation for each group, understanding that each variable is specified to follow a normal distribution with the standard deviations estimated from the data.

Table: Multilevel Regression

	Model N°1 (2016)		Model N°2 (2020)	
Parameter	Median	MAD_SD	Median	MAD_SD
(Intercept)	0.4	0.3	1.1	0.3
repvote	-1.4	0.4	-2.3	0.4
Error terms				
state (Intercept)	0.112		0.16	
educ:age (Intercept)	0.271		0.31	
educ:eth (Intercept)	0.093		0.16	
male:eth (Intercept)	0.132		0.19	
educ (Intercept)	0.179		0.23	
eth (Intercept)	0.409		0.35	

Our first multilevel regression for 2016 shows that the intercept represents the expected value on the logit scale when all predictors are zero. Using the divide-by-four rule (Gelman et al. 2006), we see that the probability of respondents believing that there should be a government health insurance plan that covers all medical expenses is $10\% \pm 7.5\%$. Nevertheless, this scenario is impractical because there could never be a state where the republican share vote would be zero. On the other hand, the median estimate for the coefficient for males is -1.4, with a standard error of 0.4. Here, it is essential to note that, as do the authors Lopez-Martin et al. in their paper "Multilevel Regression and Poststratification Case Studies", the variable repvote goes from zero to one, with zero being a state that had no Republican votes and one being a state that entirely voted for the Republican party. In this case, following the methodology of the authors mentioned above, it is decided to divide the mean coefficient by ten so that it is understood as the difference between a state that obtains 10% more in the repvote variable than another state (Lopez-Martin et al. 2022). Using the divide-by-four rule (Gelman et al. 2006) again, we can see, that controlling for the other variables, someone from a state with a 60% Republican vote share has approximately $4\% \pm 10\%$ fewer probabilities of believing that there should be a government health insurance plan that covers all medical expenses relative to another individual with the same characteristics from a state with a 40% Republican vote share.

Continuing with the 2020 multilevel regression, we can appreciate that the intercept again represents the expected value on the logit scale when all predictors are zero. As we saw above, this case is impractical because there could never be a state where the Republican share vote would be zero.

On the other hand, for the Republican vote share, the media coefficient is -2.3, with a standard error of 0.4 on the logic scale. Using the same approach towards this variable and using

the divide-by-four rule (Gelman et al. 2006), we can see that for controlling the other variables, someone from a state with a 60% Republican vote share has approximately $5.75\% \pm 10\%$ fewer probabilities of believing that there should be a government health insurance plan that covers all medical expenses relative to another individual with the same characteristics from a state with a 40% Republican vote share. In that sense, we can see an increase in the median of these coefficients, but we can conclude that this is statistically significant; this would be available for further investigation.

Across the board, states experienced an increase in public opinion concerning government healthcare, but this is not a statistically significant result. Graph 1 shows the mean point estimates as well as standard deviations for each of the 50 states. A key takeaway, in addition to the relatively large standard deviations present, is the overlapping among each observation and its counterpart. This implies that there is also an overlap in the standard error as well as their confidence intervals; due to time constraints we were unable to calculate confidence intervals in each of the states.

Mississippi experienced the lowest point increase from 0.368 in 2016 to 0.3908 in 2020, with Indiana, North Dakota, Wyoming, Texas, and Alabama all showing a less than 0.03 point increase in public opinion. This is interesting when considering that Mississippi was considered to have one of the worst responses to the COVID-19 pandemic (“2022 Scorecard on State Health System Performance”). In cases like this it would be beneficial to have data from a more recent year, like 2022. Lockdown began in March of 2020, citizens did not experience the full effects of the pandemic when the 2020 ANES survey was being conducted. Also, it is worth noting that all of the states listed are ‘Red’ states.

Michigan had the highest point increase from .4128 to .5579, following close behind with an increase in point estimates over .1, California, Connecticut, Maryland, Oregon, Vermont and Washington; California had the highest point estimate in 2020. Each of these are ‘Blue’ states which could explain why they have the highest desire for government healthcare as this aligns with the Democratic party’s platform.

In Appendix A, Maps 1 and 2 represent the change in support for government health insurance in the contiguous United States. In these maps, the regional divisions in the support of government health insurance are represented more clearly. States with the highest support are concentrated on the West Coast and Northeast, whereas states with lower levels of support are concentrated in the South and Midwest. Regional disparities in the response to COVID-19, as well as average income, in each state or region are both relevant factors in understanding how the pandemic affected support for government health insurance. However, although general trends in American politics are often influenced by the region of the country and the political polarity of individual state governments, in this case an increase in support for a government health insurance program was observed nationwide.

Conclusion

Per our research, it appears that there was not a significant change in the public opinion of those living in red states, as opposed to those living in blue states. Historically, states that are more liberal tend to vote in favor of issues such as healthcare, wage increases, etc, whereas states that lean more conservative vote against these issues. However, there is an increase in all states in relation to health insurance support, but as mentioned above, this increase is not statistically significant.

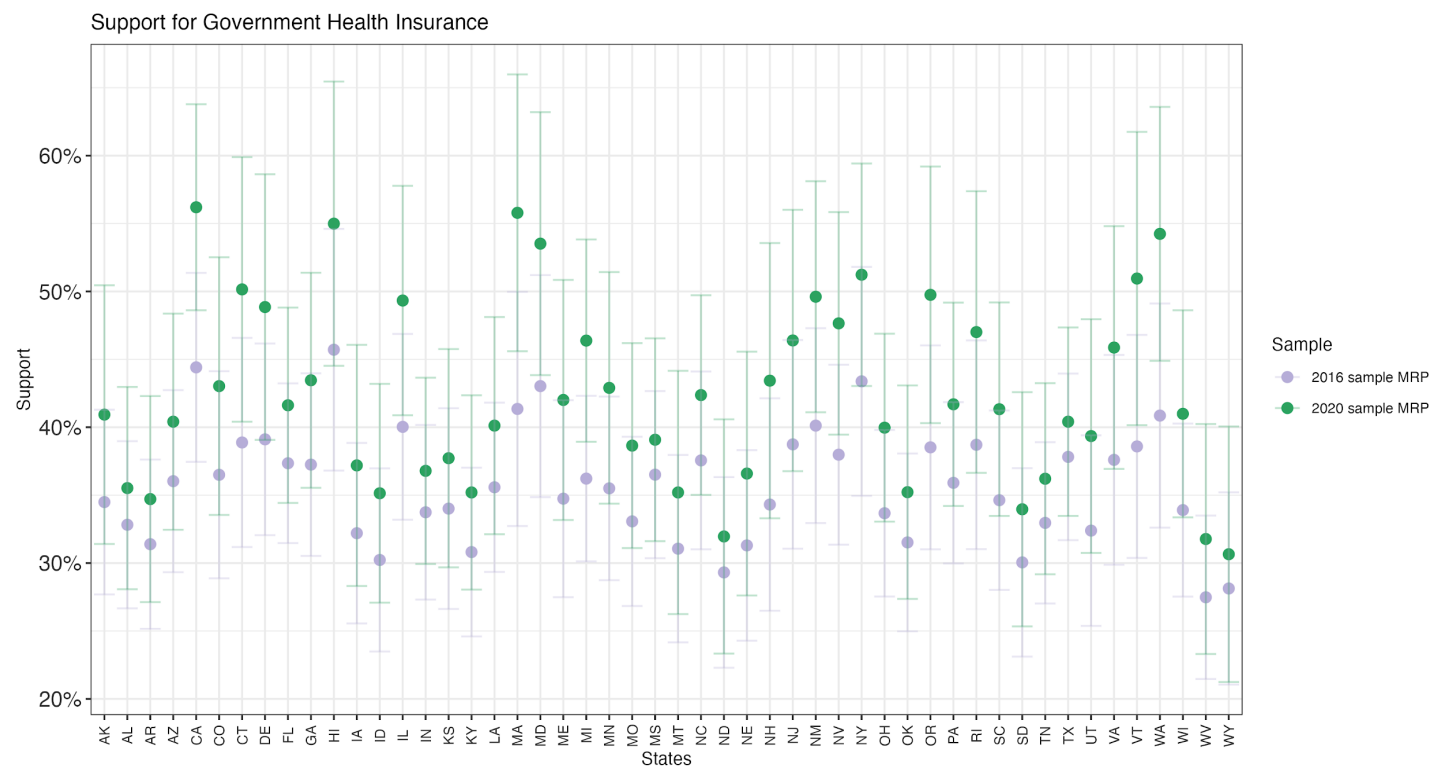
One notable observation is the efficacy of multilevel regression and poststratification in estimating public opinion within subnational populations. Population surveys are costly and time-intensive, posing challenges in obtaining specific subnational parameters and generally focusing on national parameters. This methodology, however, offers a workaround by taking advantage of national surveys. Through poststratification we can derive subnational population estimates, this approach benefits policymakers and politicians seeking crucial insights for policy implementation. Despite the manifold advantages of estimating public opinion, it remains imperative to conscientiously acknowledge and address the inherent limitations while interpreting the outcomes.

Future Works

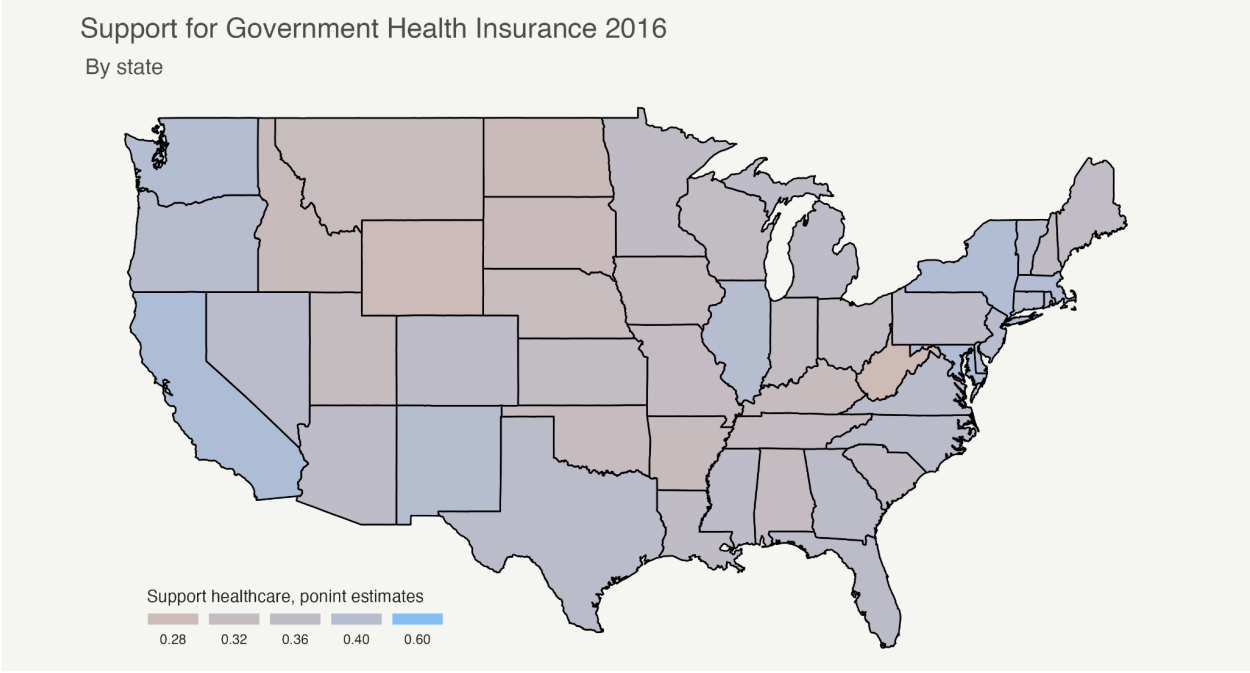
Given the short time frame for the project, we were unable to access certain datasets. Initially, we wanted to explore the Census/Integrated Public Use assessed sample and the L2 data, however we could not retrieve the data due to the IRB protocol. We were also unable to generate confidence intervals for our dataset. This would have been very helpful in showing the probability that our sample lies between two values. As aforementioned, this research falls flat by not incorporating multiple years. In our analysis, we only explored 2016 and 2020. By incorporating more years in the next iteration of this research, we would be able to see more trends and patterns that exist across the years. In the future, it would be useful to extend this research to the county-level and determine how county opinion has changed over the last five years with regard to affordable healthcare.

Appendix A: Data Visualizations

Scatter plot 1



Map 1



Map 2

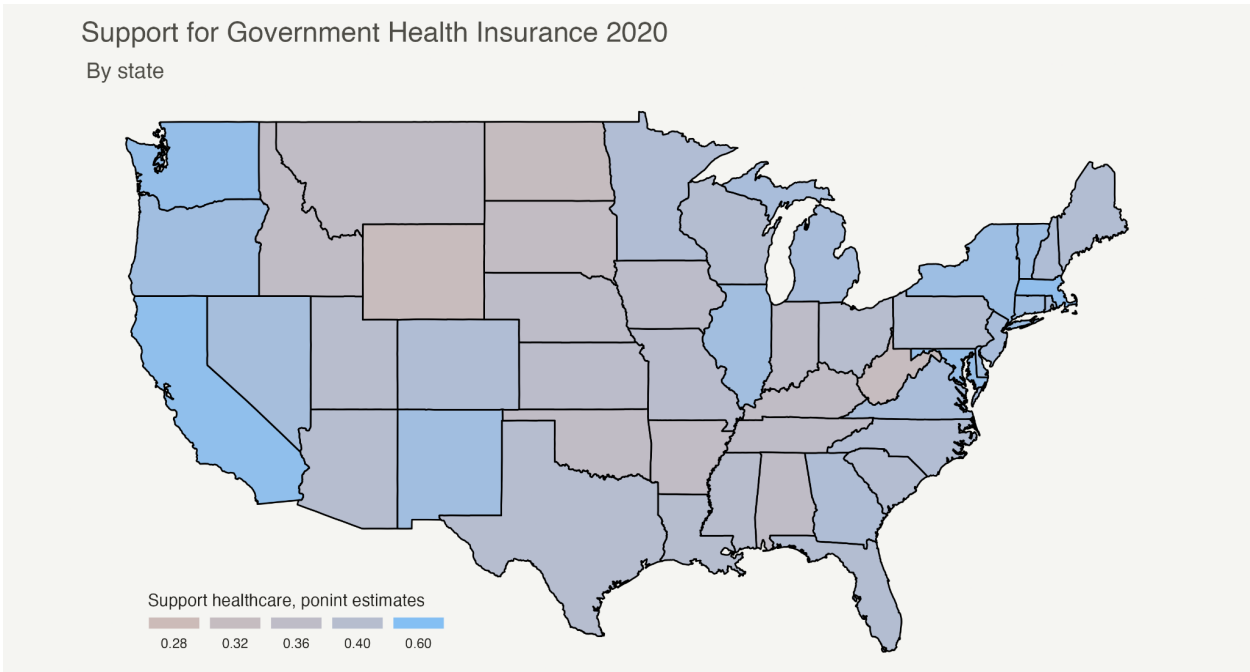


Table 1

Table N°1: ANES datasets

Variables		ANES 2016		ANES 2020	
		Mean	Std. Dev.	Mean	Std. Dev.
male		0.0	0.5	0.0	0.5
Government health insurance		0.4	0.5	0.5	0.5
		N	Pct.	N	Pct.
age	18-29	651	15.2	1003	12.1
	30-39	686	16.1	1230	14.9
	40-49	553	13.0	1076	13.0
	50-59	685	16.0	1181	14.3
	60-69	714	16.7	1404	17.0
	70+	567	13.3	1452	17.5
educ	NA	414	9.7	934	11.3
	4-year college +	1635	38.3	3647	44.0
	HS	1056	24.7	1336	16.1
	No HS	80	1.9	507	6.1
	Some college	1499	35.1	2790	33.7
eth	Black	397	9.3	726	8.8
	Hispanic	450	10.5	762	9.2
	Other	352	8.2	727	8.8
	White	3038	71.1	5963	72.0
	NA	33	0.8	102	1.2

Appendix B: Works Cited

American National Election Studies. *Time Series Cumulative Data File (1948-2020)*.

<https://electionstudies.org/data-center/anes-time-series-cumulative-data-file/>. Accessed 15 Dec. 2023.

“2022 Scorecard on State Health System Performance.” *COVID-19 | Commonwealth Fund*, 16 June 2022,

www.commonwealthfund.org/publications/scorecard/2022/jun/2022-scorecard-state-health-system-performance#:~:text=Hawaii%2C%20New%20Hampshire%2C%20and%20Washington,than%20500%20deaths%20per%20100%2C000.

Gelman, Andrew, and Jennifer Hill. 2006. *Data Analysis Using Regression and Multilevel/Hierarchical Models*. Cambridge university press.

“Health Insurance Coverage of the Total Population.” *KFF*,

<https://www.kff.org/other/state-indicator/total-population/>. Accessed 14 Dec. 2023.

Lopez-Martin, Juan, et al. “Multilevel Regression and Poststratification Case Studies.”

Bookdown, 16 May 2022, bookdown.org/jl5522/MRP-case-studies/.

Rakus, Alexandra, and Aparna Soni. “Association between State Medicaid Expansion Status and Health Outcomes during the COVID-19 Pandemic.” *Health Services Research*, vol. 57, no. 6, 2022, pp. 1332–41. *Wiley Online Library*,

<https://doi.org/10.1111/1475-6773.14044>.

U.S. Census Bureau. (n.d.). Without health care coverage in ZCTA5 90804. U.S. Department of Commerce. Retrieved September 27, 2022, from <https://data.census.gov/>

Wilper, Andrew P., et al. "A National Study of Chronic Disease Prevalence and Access to Care in Uninsured U.S. Adults." *Annals of Internal Medicine*, vol. 149, no. 3, Aug. 2008, p. 170. *DOI.org (Crossref)*, <https://doi.org/10.7326/0003-4819-149-3-200808050-00006>.